

Creative Through Constraints: How Old Mechanical Automata Fosters Innovation in Shadow Puppetry

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Figure 1: This is one of the artworks integrating automata with shadow puppetry. The work "Dream and Reality", based on "Ode to the Goddess of the Luo River", portrays the dreamlike encounter between Cao Zhi and the goddess, exploring the distance between ideal and reality. Through rotating the central gear mechanism, the installation enables vertical movement of shadow puppet characters, triggering story progression. The work presents dual aspects of the narrative through its front and back perspectives.

Abstract

While digital technologies dominate cultural heritage innovation, this paper explores an alternative approach using mechanical automata. Inspired by Yokoi's "Withered Technology" philosophy, we conducted a workshop where 33 participants created eight shadow play installations. Our findings demonstrate how mechanical constraints can foster creativity while maintaining cultural authenticity, challenging conventional assumptions about technological advancement. We provide a perspective for evaluating technology selection in cultural heritage projects and practical insights for designers working with mechanical constraints.

CCS Concepts

• **Human-centered computing** → *Empirical studies in accessibility*; • **Applied computing** → *Media arts*.

Keywords

Cultural Heritage Innovation; Shadow Puppetry; Automata; Low-Tech; Design Constraints

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1 introduction

Cultural heritage innovation often gravitates toward advanced technological solutions, with practitioners and researchers continually seeking cutting-edge tools and approaches. While this technological advancement offers new possibilities, it also presents significant challenges. The high technical threshold often prevents cultural practitioners from direct participation in the innovation process [15]. The emphasis on technological complexity can distance audiences from authentic cultural experiences [10], and the focus on technological novelty sometimes overshadows the inherent values of cultural heritage [22].

This situation prompts us to consider alternative approaches, inspired by Gunpei Yokoi's philosophy of "Withered Technology" - the strategic use of mature, well-understood technologies in novel ways [3]. Yokoi demonstrated that innovation doesn't necessarily require cutting-edge technology, but rather creative applications of established solutions [11]. This perspective suggests an intriguing possibility: could "old" technologies offer unique advantages in cultural heritage innovation?

In this paper, we define cultural heritage innovation as the process of reimagining and revitalizing traditional cultural practices through thoughtful integration with technological approaches,

while maintaining their core cultural values and authenticity. Effective cultural heritage innovation should balance three key elements: preserving the essence of cultural traditions, enhancing accessibility for contemporary audiences, and creating new expressive possibilities that extend rather than replace traditional forms. This definition moves beyond technological novelty alone, instead emphasizing how technology selection and application can foster meaningful engagement with cultural heritage while respecting its inherent characteristics and historical significance.

We explore this question through the integration of two mature technologies: traditional shadow puppetry and mechanical automata. Both represent well-documented forms with rich knowledge bases: shadow puppetry embodies centuries of artistic tradition and performance techniques [7], while automata offers established mechanical principles and crafting methods [1]. These technologies, while considered "outdated" by some standards, provide accessible yet sophisticated means for cultural expression [8].

We conducted a design workshop with 33 designers who created eight distinct shadow play installations using mechanical automata, followed by both physical and online exhibitions. Our study examines how working within defined technological boundaries can lead to creative cultural expressions while maintaining accessibility and authenticity. This research offers two key contributions:

- A new design perspective that reconsiders the potential of mature technologies in cultural heritage innovation, challenging the common assumption that newer technology necessarily leads to better outcomes
- A complete workshop process and artistic outcomes that provide design implications for technology selection in cultural heritage innovation

2 Related work

2.1 Shadow Play Digitalization and Cultural Heritage Innovation

Digital technology has become an important tool for preserving and revitalizing cultural heritage and traditional culture. Extensive efforts have been made to utilize digital technologies for the documentation, presentation, dissemination, and education of cultural heritage [6, 16, 20, 21].

In the context of shadow puppetry, digital innovation has followed several distinct approaches. One major direction focuses on developing digital shadow play systems that capture and transform traditional puppet movements, including gestures and body movements, into virtual performances [12, 24, 25]. Another stream of research builds on traditional shadow puppet patterns to enable personalized design creation [14]. For example, Yao et al. presented Lumina, a design tool that provides templates and AI-generated patterns to assist in digital shadow puppet creation, aiming to free creators from tedious tasks [23].

Some researchers have explored shadow puppetry as inspiration for digital storytelling and collaborative narratives [13, 17]. Others have attempted to break the two-dimensional performance limitations of traditional shadow play by providing immersive 3D experiences through VR/XR technologies [4, 5]. These approaches aim to expand the expressive possibilities of traditional shadow puppetry while preserving its cultural essence.

While digital technologies offer significant potential for cultural heritage innovation, both practitioners' technical barriers and researchers' over-emphasis on technological novelty present significant challenges in their effective application [15]. These challenges raise an important question: rather than constantly pursuing new technologies, could mature or "old" technologies offer unique advantages in cultural heritage innovation? This perspective leads us to explore the role of constraints in design innovation, particularly through mechanical approaches.

2.2 The Role of Constraints in Design Innovation

Research on design constraints has increasingly recognized their potential to foster creativity rather than hinder it. Biskjaer and Halskov's study of digital design practice found that constraints can serve as creative triggers, pushing designers to explore novel solutions [19]. In the context of cultural heritage, Giaccardi and Palen highlight how technological constraints can actually enhance cultural authenticity by forcing deeper engagement with traditional practices [2]. Jung and Stolterman emphasize how material constraints in digital design can lead to more thoughtful and meaningful outcomes [9].

The value of constraints has been particularly noted in cultural heritage projects. Some works [10, 18] point out how constraints can help maintain cultural authenticity while fostering innovation. However, existing research typically focuses on how constraints might guide the application of new technologies, leaving a gap in understanding how deliberately choosing and working with mature technologies might offer unique advantages in cultural heritage innovation. Our work addresses this gap by examining how mechanical automata, a mature technology with well-understood constraints, can foster new forms of cultural expression while maintaining traditional craft values.

3 Workshop Design

To investigate how mechanical constraints foster cultural innovation, we conducted a six-week design workshop with 33 undergraduate students majoring in Digital Media Arts. This workshop was carefully structured to explore the integration of mechanical automata with traditional shadow puppetry through hands-on design practice.

3.1 Participant Selection

We recruited 33 participants (aged 19-22) and organized them into eight teams. These participants possessed fundamental 2D/3D design skills and demonstrated critical thinking capabilities through their prior coursework. While many had experience in interactive design projects, none had prior experience with either automata construction or traditional shadow puppetry, making them ideal candidates for exploring innovative approaches without preconceived constraints.

3.2 Knowledge Preparation

The workshop began with a comprehensive knowledge preparation phase to ensure participants had a solid foundation in both mechanical principles and cultural context. For automata knowledge,

we traced its evolution from medieval mechanical crafts to contemporary entertainment devices, emphasizing the integration of mechanical precision with artistic expression. Participants learned fundamental mechanical principles through both theoretical explanations and hands-on demonstrations, including the use of basic mechanisms like cams, linkages, and gears. To support practical implementation, we provided training in both traditional prototyping techniques using cardboard and wood, and modern digital fabrication tools such as laser cutters and 3D printers. The shadow puppetry introduction focused on its artistic characteristics, manipulation techniques, and cultural significance, providing participants with essential context for their innovative explorations.

3.3 Design Constraints

In this workshop, we established three types of design constraints to guide the creative process, each serving as a catalyst for innovation rather than a limitation. These constraints were carefully structured to promote both technical innovation and cultural preservation while ensuring meaningful audience engagement.

Physical constraints centered on mechanical movements as the core of each installation. While teams could incorporate additional technologies as supplements, all primary movements and interactions had to be achieved through mechanical structures such as gears, cams, and linkages. With a minimum installation size of 90cm, these mechanical systems needed to form the foundation of the artistic expression. These fundamental requirements challenged teams to explore the expressive potential of mechanical movements, pushing them to develop sophisticated mechanical combinations that could effectively convey cultural narratives and create engaging experiences.

Technical constraints focused on material exploration and sustainability. Rather than using specialized or high-tech materials, teams were encouraged to innovate with readily available materials such as wood, acrylic, and recycled components. This approach emphasized the creative potential in common materials, challenging teams to discover new applications and combinations of existing resources. By limiting material choices, we aimed to demonstrate how innovation could emerge from resourceful use of accessible materials rather than relying on advanced technologies.

Interactive constraints emphasized the relationship between the mechanical system and its operators/audience. All interactions had to be manually operated through intuitive mechanisms that maintained clear cause-and-effect relationships. This requirement preserved the intimate connection between performer and puppet characteristic of traditional shadow puppetry, while ensuring operational reliability and audience engagement. Teams needed to design interactions that could be easily understood and operated while still creating meaningful cultural experiences.

These three types of constraints worked in concert to create a unique design space where innovation emerged from the creative navigation of limitations. Rather than relying on technological complexity, teams were challenged to achieve sophisticated cultural expressions through fundamental mechanical principles and sustainable material choices.

3.4 Expert Support Structure

Throughout the workshop, two expert instructors provided ongoing guidance and support. The first expert, with expertise in automata and HCI, focused on integrating mechanical and cultural elements, helping teams navigate the challenges of combining traditional shadow puppetry with mechanical innovations. The second expert, specializing in digital fabrication, offered technical support for implementation, assisting with mechanical design refinement and fabrication processes. These experts conducted regular consultation sessions with all teams, reviewing design concepts for technical feasibility and cultural appropriateness while sharing relevant case studies to inspire creative solutions.

3.5 Project Requirements

All teams were tasked with creating two key deliverables: an interactive installation and a design process portfolio. The installation requirement specified a minimum dimension of 90cm, encouraging participants to create substantial works that could offer rich cultural experiences through mechanical interactions. This physical scale requirement challenged teams to consider both structural stability and interactive engagement in their designs. The accompanying portfolio (15-20 pages) served to document each team's complete design journey, from initial concept to final implementation, capturing their decision-making process and reflections on combining traditional cultural elements with mechanical innovation.

3.6 Data Collection

Our data collection process followed multiple approaches to ensure comprehensive documentation of both the design process and outcomes:

Documentation Analysis: We systematically coded and analyzed the eight teams' design documentation, creating structured datasets that captured key aspects of their design evolution. These data were organized into comparative matrices (see the table in supplementary material) focusing on mechanical solutions, cultural interpretations, and interaction designs.

Expert Observations: Our two expert instructors maintained detailed observation records throughout the workshop. The automata and HCI expert documented teams' approaches to mechanical-cultural integration, while the fabrication expert recorded technical problem-solving strategies and material applications. These observations provided valuable insights into teams' design processes and decision-making patterns.

Exhibition Feedback: Upon completion, we organized both physical and online exhibitions of the eight installations, attracting approximately 300 visitors. Through random sampling, we collected and systematically documented audience feedback, focusing on their interactions with the installations and their perspectives on the mechanical-cultural integration. This exhibition phase provided valuable third-party perspectives on the effectiveness of different design approaches.



(a).Balance

Based on "The Farmer and the Snake", exploring the conflict between village expansion and environmental protection. Audience influences the balance through choosing "cutting trees for houses" or "protecting trees".



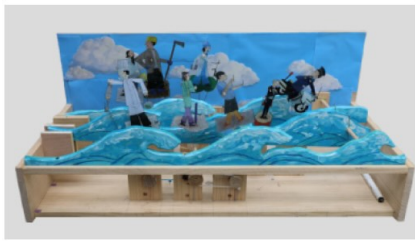
(b).Tian Nv

Adapted from "The Cowherd and the Weaver Girl", incorporating modern reflections on women's freedom and destiny.



(c).Shadow of Tang Dynasty

Based on Tang Dynasty culture, presenting three layers: Bottom: Tang Dynasty painting art. Middle: Court life scenes. Top: Tang Dynasty court banquet atmosphere



(d).The "Eight Immortals" Cross The Sea

Adapted from Chinese mythology, innovatively mapping immortals to modern professions.



(e).Dream and Reality

Based on "Ode to the Goddess of the Luo River", portraying the dreamlike encounter between Cao Zhi and the goddess, exploring the distance between ideal and reality.



(f).Utopia in Dreams

Adapted from "Peach Blossom Spring": Day: Pastoral utopia. Night: Mysterious adventure. Day-night transition for temporal dimension



(g).The Unseen Corner

Exploring marriage reality in Republican era, using traditional lacquer jewelry box form to show the duality of married life.



(h).Ecological Light

Based on 'Gold for Water', exploring environmental balance through interactive light and shadow.

Figure 2: Interactive installations (minimum dimensions 90cm × 90cm × 90cm). (a-h) 8 artworks that integrate shadow play in different ways.

4 Findings

4.1 8 Artworks

To understand how mechanical constraints foster innovation in cultural heritage design, we first analyzed the content and technical approaches of all eight installations (see figure 2). Each project demonstrated unique strategies in combining traditional shadow puppetry with mechanical movements:

"Balance" (see figure 2-a) explored the tale of "The Farmer and Snake" through an ecological development lens. The installation

cleverly utilized mechanical balance and counterweight structures, where the tilting mechanisms physically embodied the tension between urban development and environmental protection. Through mechanical movements, audiences could experience how their choices affected the equilibrium between these competing forces.

"Tian Nv" (see figure 2-b) reinterpreted the love story of "The Cowherd and Weaver Girl" through a feminist lens. Its clock-like rotating mechanism with sophisticated gear combinations enabled shadow puppets to move across multiple spatial layers, presenting stories of seven pioneering women from different historical periods.

"Shadow of Tang Dynasty" (see figure 2-c) recreated scenes of Tang Dynasty daily life through a three-tiered mechanical theater. Each layer employed distinct mechanisms - wave mechanisms for foundational scenes, gear transmissions for props, and axial rotations for character animations - creating a progressive journey through historical spaces.

"The 'Eight Immortals' Cross The Sea" (see figure 2-d) reimagined the traditional tale of deities crossing the sea by depicting the immortals as modern professionals. The precision mechanical transmission system coordinated multiple puppet movements, translating traditional characters into contemporary contexts.

"Dream and Reality" (see figure 2-e) presented the classical story of "Luo River Goddess," exploring the dreamlike encounter between Cao Zhi and the goddess. Its double-sided mechanical theater used rotational mechanisms to contrast dream and reality through shadow play.

"Utopia in Dreams" (see figure 2-f) adapted Tao Yuanming's "Peach Blossom Spring," depicting an isolated utopia. The installation used mechanical day-night transitions to contrast ideal and reality, combining automated scene changes with manual interactions.

"The Unseen Corner" (see figure 2-g) examined the dual nature of marriage traditions in the Republican era. Designed as a mechanical jewelry box, it revealed contrasting wedding ceremony scenes through coordinated mechanical movements.

"Ecological Light" (see figure 2-h) drew inspiration from the parable "A Bag of Gold for Water," focusing on environmental awareness. The mechanical structure incorporated sustainable materials to create shadow play narratives about human-nature relationships.

4.2 Qualitative Coding and Thematic Analysis

Based on our documentation and observations, we conducted a systematic analysis of all eight installations through a structured qualitative coding approach (see material in appendix). Our analysis began with an initial review of all project documentation and observation notes to identify preliminary patterns. Through iterative discussions, the research team developed a comprehensive coding framework consisting of six dimensions that captured key aspects of each installation's development and implementation.

Artwork Name and Team Members: We documented basic project information including Chinese and English titles, and team composition with their specific roles in the design process.

Background Story: We examined how each team interpreted and adapted traditional narratives, identifying their source materials and creative transformations of classical stories.

Installation Content: We analyzed the physical and mechanical composition of each work, including key structural elements, movement mechanisms, and material applications.

Core Design Features: We identified the primary innovations in each project, focusing on how mechanical constraints led to creative solutions in both technical implementation and cultural expression.

Deep Thinking: We documented teams' reflections on how automata mechanisms influenced their understanding and representation of traditional shadow puppetry, capturing their insights

about the relationship between mechanical constraints and cultural innovation.

To ensure analytical rigor, we employed researcher triangulation where team members cross-checked coding interpretations and resolved discrepancies through discussion. The coded data underwent multiple rounds of review to identify recurring patterns across projects. Similar patterns were clustered to form preliminary themes - for example, we noticed that multiple projects utilized mechanical structures to achieve spatial transformations of traditional shadow play, leading to the "Innovation Through Spatial Transformation" theme. These preliminary themes were refined through continuous comparison with the original data and team discussions. For instance, initial observations about mechanical storytelling were refined into the more specific "Mechanical Metaphor in Storytelling" theme based on how teams consistently used mechanical structures as metaphorical elements enhancing narrative expression.

4.3 Three Innovation Patterns

Our thematic analysis process was further validated by cross-referencing emerging themes with exhibition feedback, allowing us to assess their relevance and accuracy from audience perspectives. This systematic analytical approach enabled us to identify three key innovation patterns that demonstrate how mechanical constraints fostered creativity in shadow puppetry innovation.

Innovation Through Spatial Transformation

Teams demonstrated three primary approaches to transforming traditional two-dimensional shadow play into three-dimensional mechanical experiences. For instance, "Tian Nu" used multi-layered rotating mechanisms to create depth, while "Shadow of Tang Dynasty" employed a three-tiered spatial structure to progressively engage viewers. These spatial innovations expanded the traditional viewing experience while maintaining cultural authenticity.

Mechanical Metaphor in Storytelling

Teams consistently used mechanical structures not just as movement generators but as metaphorical elements that enhanced narrative expression. "Balance" exemplified this approach by using a tilting mechanism to physically embody environmental conflicts. Similarly, "Dream and Reality" employed a dual-stage mechanical system to represent the contrast between illusion and reality.

Interactive Engagement Patterns

The mechanical nature of automata created unique opportunities for audience interaction with traditional shadow play, making cultural narratives more accessible and engaging. For example, "Utopia in Dreams" revealed different story scenes through hand-cranked rotational mechanisms, while "Balance" cleverly utilized the properties of equilibrium structures to provoke audience reflection on the narrative. These interfaces preserved the tactile quality of traditional shadow puppetry while establishing meaningful connections between operators and cultural content. These findings suggest that mechanical(old-tech) constraints, rather than limiting creative expression, actually fostered innovation by:

- Pushing teams to deeply understand and creatively interpret traditional cultural elements
- Encouraging the development of novel spatial and interactive solutions

- Creating engaging human-installation relationships that enhanced cultural expression

5 Discussion

Our findings reveal several key insights about using mechanical constraints in cultural heritage innovation, while also highlighting important limitations of our approach.

5.1 Benefits of Mechanical Constraints

The accessibility of mechanical principles proved crucial in fostering innovation while maintaining cultural authenticity. Teams reported that basic mechanical structures allowed for clear task decomposition and rapid prototyping, enabling them to focus on cultural expression rather than technical problem-solving. As one participant noted,

"The straightforward mechanical principles allowed us to focus more on cultural expression rather than technical problem-solving."

The physical nature of mechanical interactions created immediate, tangible connections with audiences. Unlike complex digital interfaces, mechanical controls provided direct feedback that audiences could easily understand and engage with. This simplicity enhanced both user engagement and operational sustainability, as demonstrated by the consistent performance of all eight installations throughout the exhibition period.

5.2 Limitations and Methodological Considerations

While our study demonstrates the potential of mechanical constraints in cultural innovation, several limitations should be acknowledged. First, our participant pool consisted entirely of undergraduate design students, who may approach creative challenges differently than traditional craftspeople or professional designers. Their limited experience with traditional cultural practices might have influenced their interpretation and implementation of cultural elements.

The six-week timeframe, while sufficient for prototype development, may not have allowed for the deep cultural exploration that more extended engagement might provide. Additionally, the academic setting of the workshop may have created an artificial environment that differs from real-world cultural heritage innovation contexts.

5.3 Design Implications

Our findings suggest three key implications for cultural heritage innovation. First, technology selection should prioritize compatibility with cultural practices over technical sophistication. Second, defined constraints can serve as catalysts for innovation rather than limitations. Finally, mechanical approaches can create more accessible and sustainable paths for cultural heritage preservation.

These insights challenge the common assumption that advanced technology necessarily leads to better cultural innovation outcomes. Instead, they suggest that thoughtful application of mature technologies, combined with clear creative constraints, can effectively

support cultural heritage innovation while maintaining authenticity and accessibility.

6 Conclusion

This research demonstrates the unique value of traditional technologies in cultural heritage innovation. Through our exploration of automata-based design, we reveal how "low-tech" approaches can effectively bridge technical accessibility and cultural authenticity. Our findings suggest that the success of technology-culture integration depends not on technological advancement, but on achieving appropriate alignment between technical constraints and cultural expression. The mechanical nature of automata created unique opportunities for innovation while maintaining cultural authenticity, challenging the assumption that newer technology necessarily leads to better outcomes. Future research could explore how this framework might apply to other traditional technologies and cultural forms, expanding our understanding of constraint-driven innovation in cultural heritage design.

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